Luis Thomas

Observatoire de Genève
Université de Genève, Geneva, Switzerland
luis.thomas@unige.ch | Scholar | ORCID

Employment history

Postdoctoral Researcher and CHEOPS GTO Support Scientist from Nov 2026
Université de Genève, Geneva, Switzerland

Postdoctoral Researcher Jul 2026–Oct 2026
Max-Planck Institute for Extraterrestrial Physics, Munich, Germany

Education

Ph.D. Astronomy May 2022–Jun 2026
University Observatory Munich, Germany
Thesis: *RV follow-up of TESS planet candidates with the MaHPS spectrograph*
Advisor: Dr. Roberto Saglia
Grade: Summa Cum Laude (0.7)

M.Sc. Planetary Science Oct 2020–Oct 2021
University College London, UK
Thesis: *Detecting helium in exoplanet atmospheres with the Hubble Space Telescope*
Advisor: Dr. Angelos Tsiaras

B.Sc. Physics Oct 2017–Sep 2020
Ludwig-Maximilians University, Germany
Thesis: *Detecting exoplanets in the habitable zone around M-dwarfs*
Advisor: Dr. Roberto Saglia

Publications

Lead Author

5. **Thomas, L.**, Nielsen, L. D., Cridland, A., et al. (2026), *A&A* accepted
   Companion architectures of sub-Saturns: Distinct migration pathways across the Neptunian landscape
4. **Thomas, L.**, Nielsen, L. D., Kellermann, H., et al. (2026), *A&A* 712 A157
   TOI-2147 b and TOI-6019 b: Two eccentric warm Jupiters detected and characterized with TESS and MaHPS
3. **Thomas, L.**, Nielsen, L. D., Acuña-Aguirre, L., et al. (2025), *A&A* 703 A221
   Evidence of a gap in the envelope mass fraction of sub-Saturns
2. **Thomas, L.**, Hébrard, G., Kellermann, H., et al. (2025), *A&A* 694 A143
   TOI-5108 b and TOI 5786 b: Two transiting sub-Saturns detected and characterized with TESS, MaHPS and SOPHIE
1. **Thomas, L.**, Saglia, R., Pasquini, L., et al. (2024), *A&A* 686 A19
   Search for giant planets in M67 V: a warm Jupiter orbiting the turn-off star S1429

Co-Author

5. Almenara, J., Castro-González, A., Mardling, R., et al. (incl. **Thomas, L.**) (2026), *A&A* 714 A190
   HD 148797: A bright F-type star with two moderate-period low-density sub-Jovian planets

4. Grouffal, S., Santerne, A., Bourrier, V., et al. (incl. **Thomas, L.**) (2025), *A&A* 701 A173, 2025
   The star HIP 41378 potentially misaligned with its cohort of long-period planets

3. Ehrhardt, J., **Thomas, L.**, Kellermann, H., et al. (2024), *A&A* 692 A220
   Confirmation of four hot Jupiters detected by TESS using follow-up spectroscopy from MaHPS at Wendelstein together with NEID and TRES

2. Korth, J., Chaturvedi, P., Parviainen, H., et al. (incl. **Thomas, L.**) (2024), *ApJL* 971 L28
   TOI-1408: Discovery and Photodynamical Modeling of a Small Inner Companion to a Hot Jupiter Revealed by Transit Timing Variations

1. Edwards, B., Changeat, Q., Tsiaras, A., et al. (incl. **Thomas, L.**) (2023), *AJ* 166 158
   Characterising a World Within the Hot Neptune Desert: Transit Observations of LTT 9779 b with HST WFC3

Teaching & Mentorship

Teaching

Basic Lab Exoplanet Detection | Supervisor 2024 – 2026
Basic Lab Exoplanet Detection | Redesign and update of the lab manual May 2025

Student Mentoring

Paul Lotz | BSc Thesis Supervisor 2026
Mael Minig | BSc Thesis Supervisor 2026
Jebraan Mudholkar | MSc Thesis Co-supervisor since 2025
Yutong Liu | MSc Thesis Co-supervisor 2025 – 2026
Juliana Ehrhardt | MSc Thesis Supervisor 2022 – 2024
Laura Schöller | MSc Thesis Supervisor 2023 – 2024
Nicolas Dowling | MSc Thesis Co-supervisor 2022 – 2023
Khushi Marlecha | MSc Thesis Co-supervisor 2022 – 2023

Selected Talks & Posters

The Formation, Life and Fate of Exo-Neptunian Systems (Workshop), Leiden, NL Sep 2026
Companion architectures of sub-Saturns; **contributed talk**.

Multiple Planet Systems (Conference), Sofia, Bulgaria Sep 2026
Planet Multiplicity Across the Neptune Desert, Ridge, and Savanna; **contributed talk**.

Exocoffee (Journal Club), MPIA, Heidelberg, Germany Jul 2026
Evidence of a gap in the envelope mass fraction of sub-Saturns; **invited seminar**.

Astrophysikalisches Kolloquium (Colloquium), Georg-August-Universität, Göttingen, Germany Jul 2026
The Sub-Saturn Population: Interior Structure and System Architecture as Probes of Planet Formation; **invited seminar**.

ABC forum (Seminar), Astrobiology Center, Tokyo, Japan May 2026
The Sub-Saturn Population: Interior Structure and System Architecture as Probes of Planet Formation; **invited seminar**.

Star-Planet Exchange (SPEX) (Seminar), ESO Garching, Germany Dec 2025
Exploring the formation of warm sub-Saturns through interior structure modeling; **invited seminar**.

The Fate of Neptunian Worlds (Conference), Warwick, UK Sep 2025

Exploring the formation of warm sub-Saturns through interior structure modeling; **contributed talk**.

EPRV 6 (Conference), Porto, Portugal Jun 2025
Exoplanet Detection and mass measurement with the MaHPS spectrograph at Wendelstein; **poster**.

Austrian early career conference (Conference), Salzburg, Austria Mar 2024
Radial Velocity Measurements with the MaHPS spectrograph on the Wendelstein Observatory; **contributed talk**.

Exoplanets by the Lake I (Summer school), Munich, Germany Aug 2023
Radial Velocity Measurements with the MaHPS spectrograph on the Wendelstein Observatory; **contributed talk**.

Observing & Awarded telescope time

PI • HARPS • 9.5 hrs Populating the runaway accretion gap (Prog. 117.2A28)
Co-I • ESPRESSO • 122.5 hrs Probing Planetary Evolution: Masses of Young Transiting Exoplanets with ESPRESSO (Prog. 117.2AJ3)
Co-I • ESPRESSO • 28 hrs Planet evolution in- and around the desert: measuring masses of the young Neptunes orbiting TOI-942 (Prog. 114.277K)

Wendelstein: 52 nights as the sole on-site observer at Wendelstein Observatory. Responsible for the execution of science programs on the 2.1 m and 43 cm telescopes as well as telescope operating and instrument maintenance.

Service & Outreach

Scientific Referee
Astronomy and Astrophysics

Guided Tours at the Wendelstein Observatory
Giving guided tours to the general public of the observatory at Mount Wendelstein. Explaining the science programs as well as the available telescopes and instruments.

Techmax 40: Exoplaneten
Contribution to the Techmax issue on exoplanets, a booklet by the Max-Planck Society to promote science topics.